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Original Report

Factors associated with radiation therapy incidents in a large academic institution

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Abstract

Background: This study evaluated factors associated with radiation therapy (RT) planning and delivery incidents at a large academic institution.

Methods and materials: The RT incidents (including near-misses) were recorded using an electronic incident reporting system from April 1, 2011 to April 30, 2013. Each incident's origin was categorized according to the step in the treatment process (simulation, physician prescription, treatment planning, scheduling, treatment delivery, and other) in which it occurred. The incident database was linked to the RT delivery (record and verify) database to evaluate the effect of various factors on the rate of RT incidents.

Results: There were 189 reported RT incidents (including near-misses) among 326,448 fractions, of which there were 70 (37%) treatment planning incidents and 56 (30%) treatment delivery incidents. The rates of total incidents, planning incidents, and delivery incidents were 136.0, 50.4, and 40.3 per 10,000 patients, respectively. Logistic multivariate analysis showed that fewer work days from plan approval to treatment start, fewer fractions, higher number of prescription items, and longer beam duration were significantly associated with radiation planning incidents. Multivariate analysis also showed that first day of treatment, fewer fractions, higher number of prescription items, and longer beam duration were significantly associated with treatment delivery incidents; intensity modulated radiation therapy was associated with a lower rate of treatment delivery incidents.

Conclusions: More complicated radiation plans, fewer fractions, first day of treatment, and rushed processes were associated with higher risk of RT incidents. We hope that a national incident reporting database will lead to greater understanding of factors influencing the rate of RT incidents.

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Introduction

In 2001, the Institute of Medicine published a landmark report outlining the 6 aims for improvement in the quality of health care.[1] One of these aims was that health care should be "Safe - avoiding injuries to patients from the care that is intended to help them." Radiation therapy (RT) delivery requires a variety of providers (physicians, physicists, dosimetrists, and therapists) performing complex radiation planning and delivery processes that rely on increasingly complex technology and human-computer interfaces.[2] These factors increase the potential for errors in RT. Recent newspaper reports have highlighted instances of RT errors and overdoses, some leading to serious patient injuries[3-5] which led to a congressional hearing in the United States to investigate legislation to improve patient safety.[6]

Single institutions have reported the rate of treatment incidents in RT to be 0.1 to 4.7 per 100 patients.[7-16] A recent study showed that treatment planning represented the most frequent origin of all reported incidents, while treatment delivery represented the most frequent origin of actual incidents.[7] While little is known about factors that influence RT planning incidents, various factors have been identified that contribute to RT treatment delivery incidents, including treatments requiring a manual process,[10] using accessory devices,[11] not using portal imaging,[14] and the lack of record and verify systems.[9,13] The purpose of this study was to further elucidate the factors associated with RT treatment planning and delivery incidents in a large academic institution.

Methods and materials

Radiation therapy incidents

The RT incidents (including near-misses) were reported at the time of the event using an online quality improvement incident reporting system as previously described.[8] The incident origin was categorized according to the step in the treatment process in which it occurred (simulation, physician prescription, treatment planning, scheduling, treatment delivery, and other). For example, treatment planning incidents included incidents such as sending an incorrect digitally reconstructed radiograph or mislabeling fields, while treatment delivery incidents included incidents such as incorrect isocenter localization or not using prescribed bolus. Each incident was classified into 1 of the following 4 categories based on severity: level I (wrong patient, wrong type of radiation, wrong energy, wrong site, or total dose differing by more than 20% of the prescribed dose); level II (wrong field, wrong localization, wrong timing, total dose differing by more than 10% of the prescribed dose, or other serious event); level III (any other event reaching the patient); and level IV (near-miss). All incidents were reviewed monthly by the Quality and Safety Council, a multidisciplinary team of physicians, physicists, dosimetrists, and therapists, which formulates appropriate corrective actions and recommendations to prevent future incidents. All incidents were also reviewed at monthly faculty quality assurance meetings. Moreover, all new definitively treated patients were reviewed at weekly chart rounds, both at the main center and at the satellites.

Study design

For this analysis, information was obtained from the departmental radiation oncology quality improvement database regarding RT patient incidents among patients whose treatment was completed from April 1, 2011 to April 30, 2013. These incidents were expressed as a proportion of treated patients and treatment fractions, based on radiation oncology departmental records. Patients treated with separate courses of radiation therapy during this time period were counted as multiple patients. The incident database was linked to the RT delivery (record and verify) database which contains information on date of treatment, radiation beam duration, radiation prescription, treatment modality, technique, number of fractions, and other treatment details. Patients were only included if they completed their entire course of treatment during the inclusion dates. The number of prescription items was defined as the number of items in the RT prescription and was used as a surrogate for plan complexity. For example, a breast RT treatment might consist of the following 3 items in the RT prescription: (1) medial and lateral tangent fields; (2) internal mammary chain field; and (3) supraclavicular field. The radiation oncology quality improvement database was also linked to a departmental workflow tracking database that includes information on date and time of simulation, physician submitted contours, plan approval by physician and treatment start. Work days excluded weekends and institutional holidays. The study was deemed to be exempt from review by the institutional review board as a quality improvement project.

Statistics

The Pearson χ^2 test was used to assess measures of univariate association for categoric variables and logistic regression was used to assess measures of univariate association for continuous variables. A P value $<.05$ was considered to be statistically significant. Statistical tests were based on a 2-sided significance level. Logistic regression was used to assess the effect of patient and treatment characteristics on the endpoint. The 2 independent endpoints for these analyses were the following: (1) treatment planning; and (2) treatment delivery incidents.

All variables were assessed on a univariate basis and factors with a significance of 0.10 or less were assessed for multivariate analysis using backwards elimination. Data analysis was performed using Stata/IC 12.0 (StataCorp, College Station, TX) and SPSS 21 (IBM, New York, NY) statistical software.

Table 1 Univariate analysis on factors associated with treatment planning incidents

Variable		Total fractions	Treatment planning incident			*P* value
			No	Yes	Rate per 10,000 fractions	
All patients		326,448	326,378	70	2.1	
Age at treatment start						
	≤18	9648	9642	6	6.2	.006
	>18	316,800	316,736	64	2.0	
Sex						
	Male	165,794	165,763	31	1.9	.28
	Female	160,654	160,615	39	2.4	
Work days from simulation to start						
	<2	27,464	27,453	11	4.0	.03 [a]
	≥2	224,570	224,527	43	1.9	
	Unknown	74,414				
Work days from plan approval to start						
	<2	38,234	38,223	11	2.9	<.001 [a]
	≥2	214,607	214,568	39	1.8	
	Unknown	73,607				
Total no. of fractions delivered						
	≤10	53,515	53,488	27	5.0	<.001 [a]
	10-25	116,635	116,614	21	1.8	
	26-35	121,796	121,778	18	1.5	
	36+	34,502	34,498	4	1.2	
No. of prescription items						
	1	251,326	251,293	33	1.3	<.001 [a]
	2	39,221	39,204	17	4.3	
	3	14,311	14,303	8	5.6	
	4+	21,590	21,578	12	5.6	
3-dimensional radiation therapy [b]						
	Yes	162,716	162,673	43	2.6	.06
	No	163,732	163,705	27	1.6	
Intensity modulated radiation therapy [b]						
	Yes	132,364	132,346	18	1.4	.01
	No	194,084	194,032	52	2.7	
Stereotactic treatment [b]						
	Yes	3278	3277	1	3.1	.72
	No	323,170	323,101	69	2.1	
Photons [b]						
	Yes	255,085	255,039	46	1.8	.01
	No	71,363	71,339	24	3.4	
Electrons [b]						
	Yes	49,418	49,397	21	4.2	.001
	No	277,030	276,981	49	1.8	
Protons [b]						
	Yes	53,057	53,042	15	2.8	.24
	No	273,391	273,336	55	2.0	
Radiation beam duration (min)						
	≤2.5	48,370	48,368	2	0.4	<.001 [a]
	2.51-6	81,686	81,679	7	0.9	
	6.1-10	79,269	79,262	7	0.9	
	>10	90,042	90,004	38	4.2	
	Unknown	27,081				

[a] These *P* values represent results from a univariate logistic regression model.
[b] Each modality used during each fraction was included, thus 1 fraction could consist of multiple modalities.

Results

There were 189 reported RT incidents among 326,448 fractions and 13,899 patients, of which there were 70 (37%) treatment planning incidents and 56 (30%) treatment delivery incidents. The remaining 63 (33%) incidents happened at other steps of the process. The overall rate of RT incidents was 136.0 per 10,000 patients.

Treatment planning incidents

Treatment planning incidents included 0 level I (most severe), 0 level II, 16 level III (least severe), and 54 level IV (near-miss) incidents. The rate of treatment planning incidents was 50.4 per 10,000 patients and 2.1 per 10,000 fractions. Results of univariate analysis on factors associated with treatment planning incidents are displayed in Table 1. Factors significantly associated with a higher rate of treatment planning incidents on univariate analysis included younger age, fewer work days from simulation to treatment start, fewer work days from plan approval by physician to treatment start, fewer total number of fractions, greater number of prescription items, longer radiation beam duration, and electron treatment. The use of intensity modulated radiation therapy (IMRT) and photon treatment were associated with a lower rate of treatment planning incidents on univariate analysis.

Logistic multivariate analysis showed that fewer work days from plan approval to treatment start, fewer total number of fractions, greater number of prescription items, and longer radiation beam duration were independently associated with a higher rate of RT planning incidents (Table 2).

Treatment delivery incidents

Treatment delivery incidents included 0 level I (most severe), 18 level II, 33 level III (least severe), and 5 level IV (near-miss) incidents. The rate of RT treatment delivery incidents was 40.3 per 10,000 patients and 1.7 per 10,000 fractions. Results of univariate analysis on factors associated with treatment delivery incidents are displayed in Table 3. Factors significantly associated with a higher rate of treatment delivery incidents on univariate analysis included first day of treatment, fewer total fractions, greater number of prescription items, 3-dimensional RT, electron treatment, and longer radiation beam duration. The use of IMRT and photon treatment was associated with a lower rate of treatment delivery incidents on univariate analysis.

Logistic multivariate analysis showed that the first day of treatment, fewer total fractions, higher number of prescription items, and longer radiation beam duration, were independently associated with a higher rate of RT treatment delivery incidents; IMRT was associated with a lower rate of treatment delivery incidents (Table 4).

Table 2 Multivariate logistic regression on factors associated with treatment planning incidents

Variable		OR	95% CI	*P* value
Work days from plan approval to start				
	Continuous	0.97	0.95-0.99	.006
Total no. of fractions delivered				
	Continuous	0.96	0.92-0.999	.045
No. of prescription items				
	Continuous	1.47	1.15-1.73	.001
Radiation beam duration (min)				
	Continuous	1.04	1.03-1.05	.001

CI, confidence interval; OR, odds ratio.

Discussion

This study analyzed factors associated with RT treatment planning and delivery incidents at a large academic institution. We found that less time from plan approval to start, fewer fractions of RT, more prescription items, and longer radiation beam duration were independently associated with an increased risk of RT planning incidents, on multivariate analysis. Moreover, multivariate analysis showed an increased risk of RT treatment delivery incidents on the first day of treatment, when fewer fractions were delivered, when IMRT was not used, with more prescription items, and with increased radiation beam duration.

The RT planning is an intensive, detail-oriented, and complex process. Recent studies have reported that physician performance during RT planning tasks declined with increased workload levels and cross-coverage conditions.[17] Our study suggests that radiation planning performance declines with rushed processes; shorter time between physician plan approval and treatment start was associated with a higher rate of RT planning incidents. Patients treated with fewer total fractions also had a higher rate of RT planning incidents; fewer fractions may be a surrogate for palliative treatments, which are often rushed. Furthermore, more complex treatments appear to increase the rate of planning incidents. More prescription items and longer radiation beam duration could signify a more complicated treatment planning process and thus more potential for RT planning incidents.

Radiation treatment delivery is a complex process downstream from physician prescription, RT planning, and physics review. There appears to be a vulnerability associated with the first day of treatment, with a higher rate of delivery incidents that day. This vulnerability may arise because therapists are encountering a specific patient or

Table 3 Univariate analysis on factors associated with treatment delivery incidents

Variable		Total fractions	Treatment delivery incident			*P* value
			No	Yes	Rate per 10,000 fractions	
All patients		326,448	326,392	56	1.7	
Age at treatment start						
	≤18	9,648	9,647	1	1.0	.61
	>18	316,800	316,745	55	1.7	
Sex						
	Male	165,794	165,768	26	1.6	.51
	Female	160,654	160,624	30	1.9	
Work days from simulation to start						
	<2	27,464	27,454	10	3.6	.07 [a]
	≥2	224,570	224,536	34	1.5	
	Unknown	74,414				
Work days from plan approval to start						
	<2	38,234	38,226	8	2.1	.17 [a]
	≥2	214,607	214,574	33	1.5	
	Unknown	73,607				
First day of treatment						
	Yes	25,482	25,457	25	9.8	<.001
	No	300,966	300,935	31	1.0	
Total no. of fractions delivered						
	≤10	53,515	53,494	21	3.9	<.001 [a]
	10-25	116,635	116,617	18	1.5	
	26-35	121,796	121,780	16	1.3	
	36+	34,502	34,501	1	0.3	
No. of prescription items						
	1	251,326	251,301	25	1.0	<.001 [a]
	2	39,221	39,202	19	4.8	
	3	14,311	14,305	6	4.2	
	4+	21,590	21,584	6	2.8	
3-dimensional radiation therapy [b]						
	Yes	162,716	162,675	41	2.5	<.001
	No	163,732	163,717	15	0.9	
Intensity modulated radiation therapy [b]						
	Yes	132,364	132,353	11	0.8	.001
	No	194,084	194,039	45	2.3	
Stereotactic treatment [b]						
	Yes	3,278	3,277	1	3.1	.56
	No	323,170	323,115	55	1.7	
Photons [b]						
	Yes	255,085	255,048	37	1.5	.03
	No	71,363	71,344	19	2.7	
Electrons [b]						
	Yes	49,418	49,401	17	3.4	.001
	No	277,030	276,991	39	1.4	
Protons [b]						
	Yes	53,057	53,048	9	1.7	.97
	No	273,391	273,344	47	1.7	
Radiation beam duration (min)						
	≤2.5	48,370	48,366	4	0.8	<.001 [a]
	2.51-6	81,686	81,677	9	1.1	
	6.1-10	79,269	79,266	3	0.4	
	>10	90,042	90,015	27	3.0	
	Unknown	27,081				
Treatment time						
	8AM-5PM	256,109	256,066	43	1.7	.76
	After hours	70,339	70,326	13	1.8	

[a] These *P* values represent results from a univariate logistic regression model.
[b] Each modality used during each fraction was included, thus 1 fraction could consist of multiple modalities.

Table 4 Multivariate logistic regression on factors associated with treatment delivery incidents

Variable		OR	95% CI	*P* value
First day of treatment				
	No	1		
	Yes	3.47	1.75-6.91	<.001
Total no. of fractions delivered				
	Continuous	0.96	0.93-0.99	.03
Intensity modulated radiation therapy				
	Yes	1		
	No	2.24	1.02-4.90	.04
No. of prescription items				
	Continuous	1.27	1.04-1.57	.02
Radiation beam duration (min)				
	Continuous	1.03	1.02-1.04	<.001

CI, confidence interval; OR, odds ratio.

treatment plan for the first time, or because certain incidents, such as incorrect patient positioning that is later detected by physicians on port films, inherently occur more frequently on the first day of treatment. Similar to that for radiation planning, increased treatment complexity may lead to a higher risk for RT delivery incidents; increased beam duration and increased number of prescription items were associated with higher rate of delivery incidents. On the other hand, advances in technology and more automated processes may help lower the risk of errors.[12] In our study, IMRT was associated with a lower rate of RT delivery incidents. Patient-related factors may also affect the risk of RT delivery incidents. We found a higher rate of delivery errors in patients treated with fewer fractions, who are likely the patients undergoing palliative treatments.

An understanding of factors associated with RT planning and delivery incidents may help us develop interventions to reduce the risk of these incidents. For example, policies such as allowing adequate time between physician plan approval and patient start could be formulated to prevent rushed processes. Staffing models, such as having additional staff on the first day of treatment, could be developed based on these findings. While many factors that increase the risk of incidents may be unavoidable, cognizance of such factors may help us focus more on avoiding errors when those factors are present.

This study has certain limitations which need to be addressed. All incidents were self-reported; thus, the current analysis could underestimate the number of incidents. Reporting bias may be evident as our department consists of multiple services, physical locations, and providers. In addition, certain patient factors could have contributed to the rate of RT incidents which we could not account for in the current analysis; such patient factors may include performance status, inpatient hospitalization, comorbidities, tumor stage, and symptom burden. There are also other treatment-related and human factors that could not be accounted for in this analysis, including provider experience and training and machine specifications. The majority of incidents in this study were either near-misses or incidents with low severity that are unlikely to cause patient harm. It remains unknown whether the factors identified in this study are also associated with incidents of greater severity that may cause patient harm. Finally, this study was conducted on incidents in a single, large academic department. Additional studies are warranted to evaluate whether these findings are generalizable to other institutions and settings.

Over the last century, RT has undergone tremendous technologic advancement. Treatment planning has moved from 2-dimensional planning with x-ray imaging and hand calculations to 3-dimensional computed tomographic-based planning with computer-assisted calculations.[18] In addition, innovations in image guidance assure proper daily target coverage while minimizing dose to adjacent structures.[19] More recent advancements such as proton therapy have rapidly been adapted in the United States. With these advances and their inherit complexities, efforts to learn from and minimize RT incidents are underway. The American Society for Radiation Oncology has formed a patient safety organization called the Radiation Oncology Incident Learning System (RO-ILS) in order to improve the safety and quality of radiation oncology, with a rollout expected in early 2014.[20] Given its patient safety organization status, RO-ILS will be a secure, nonpunitive avenue to report incidents and near-misses in order for the radiation oncology community to develop safer practices. Until widespread participation in this program, single institution data regarding factors contributing to RT incidents will remain important.

In summary, more complicated radiation plans, fewer fractions of RT, first day of treatment, and rushed processes increased the risk of RT incidents. A national incidence reporting database may lead to greater understanding of the factors influencing the rate of RT incidents.

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